

Exercises 8.D

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Exercise 1. Find the characteristic polynomial and the minimal polynomial of the operator N in Example 8.53.

Proof. From Exercise 8.53, we have that $N \in \mathcal{L}(\mathbf{F}^4)$ is defined by

$$N(z_1, z_2, z_3, z_4) = (0, z_1, z_2, z_3).$$

Since the only eigenvalue of N is 0, it follows immediately that the characteristic polynomial of N is

$$p(z) = z^4.$$

Thus the only possibilities for the minimal polynomial are z , z^2 , z^3 , and z^4 . Clearly T itself is not the zero map, so we must start our search with z^2 . The following calculations are simple:

$$N^2(z_1, z_2, z_3, z_4) = (0, 0, z_1, z_2),$$

$$N^3(z_1, z_2, z_3, z_4) = (0, 0, 0, z_1),$$

$$N^4(z_1, z_2, z_3, z_4) = (0, 0, 0, 0),$$

so in this case N^4 is in fact the lowest power of N to be zero, and thus z^4 is the minimal polynomial of T . $\square$

Exercise 2. Find the characteristic polynomial and the minimal polynomial of the operator N in Example 8.54.

Proof. From Exercise 8.53, we have that $N \in \mathcal{L}(\mathbf{F}^4)$ is defined by

$$N(z_1, z_2, z_3, z_4, z_5, z_6) = (0, z_1, z_2, z_3).$$

$\square$